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Bihar Engineering University, Patna

B.Tech 1st Semester Exam., 2019 (New Course)

Time: 3 hours

Code: 100105 ***Subject: Workshop Manufacturing Practices*** ***Full Marks: 70***

Instructions:-

(i) The marks are indicated in the right-hand margin.
(ii) There are NINE questions in this paper.
(iii) Attempt FIVE questions in all.
(iv) Question No. 1 is compulsory.

Q.1 Choose the correct answer of the following (any seven) : **[2×7=14]**

(a) A zinc diffusion process is called
- (i) galvanizing
- (ii) anodizing
- (iii) parkerizing
- (iv) sherardizing

(b) The type of file used for a woodwork is
- (i) single-cut file
- (ii) double-cut file
- (iii) rasp-cut file
- (iv) Any one of the above

(c) A taper provided on the pattern for its easy and clean withdrawal from the mould is known as
- (i) machining allowance
- (ii) draft allowance
- (iii) shrinkage allowance
- (iv) distortion allowance

(d) In arc welding, the electric arc is produced between the work and the electrode by
- (i) voltage
- (ii) flow of current
- (iii) contact resistance
- (iv) All of the above

(e) In a centrifugal casting method
- (i) core is made of sand
- (ii) core is made of ferrous metal
- (iii) core is made of non-ferrous metal
- (iv) no core is used

(f) The draft or taper allowance on casting is generally
- (i) 1 to 2 mm/m
- (ii) 2 to 5 mm/m
- (iii) 5 to 10 mm/m
- (iv) 10 to 15 mm/m

(g) Which of the following welding processes uses non-consumable electrodes?
- (i) TIG welding
- (ii) MIG welding
- (iii) Manual arc welding
- (iv) Submerged arc welding

(h) In a bilateral system of tolerance, the tolerance is allowed on
- (i) one side of the actual size
- (ii) one side of the nominal size
- (iii) both sides of the actual size
- (iv) both sides of the nominal size

(i) The temperature at which the new grains are formed in the metal is called
- (i) lower critical temperature
- (ii) upper critical temperature
- (iii) eutectic temperature
- (iv) recrystallization temperature

(j) In sheet metal blanking, shear is provided on punches and dies so that
- (i) press load is reduced
- (ii) good cut edge is obtained
- (iii) warping of sheet is minimized
- (iv) cut blanks are straight

Q.2 Solve both questions : **[7+7=14]**

(a) Draw a neat sketch of metal jack plane and name its parts.

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(b) Write the specific use of (i) spokeshave and (ii) router plane.

Q.3 Solve both questions : **[7+7=14]**

(a) List out the different types of files. Write the use of any three types of files with sketches.

(b) Describe the working of spring hammer with neat sketch.

Q.4 Solve both questions : **[7+7=14]**

(a) What is tap? Explain the different types of tap that are used in fitting with neat sketch.

(b) List out the forging operations. Explain any four forging operations' with a suitable diagram.

Q.5 Solve this question : **[14]**

(a) List any four materials used for pattern making.

(b) Explain any four properties of moulding sand.

(c) What is meant by sand binder? Name the types of sand binders.

Q.6 Solve both questions : **[6+8=14]**

(a) Write four advantages of hot working process. Explain the principle of hot rolling.

(b) Explain the following hot working processes : (i) Extrusion (ii) Drawing

Q.7 Solve both questions : **[8+6=14]**

(a) Explain any four operations performed on lathe machine with neat sketch.

(b) Write a short note on sine bar.

Q.8 Solve both questions : **[6+8=14]**

(a) Write the specific application of any three measuring tools in carpentry. List out the different holding tools used in carpentry.

(b) Explain the following handsaws used in carpentry with neat sketch : (i) Rip saw (ii) Bow saw (iii) Dovetail saw (iv) Keyhole saw

Q.9 Solve both questions : **[6+8=14]**

(a) What is the minimum flange length in sheet metal bend?

(b) Draw and explain four types of seam in sheet metal operation.

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